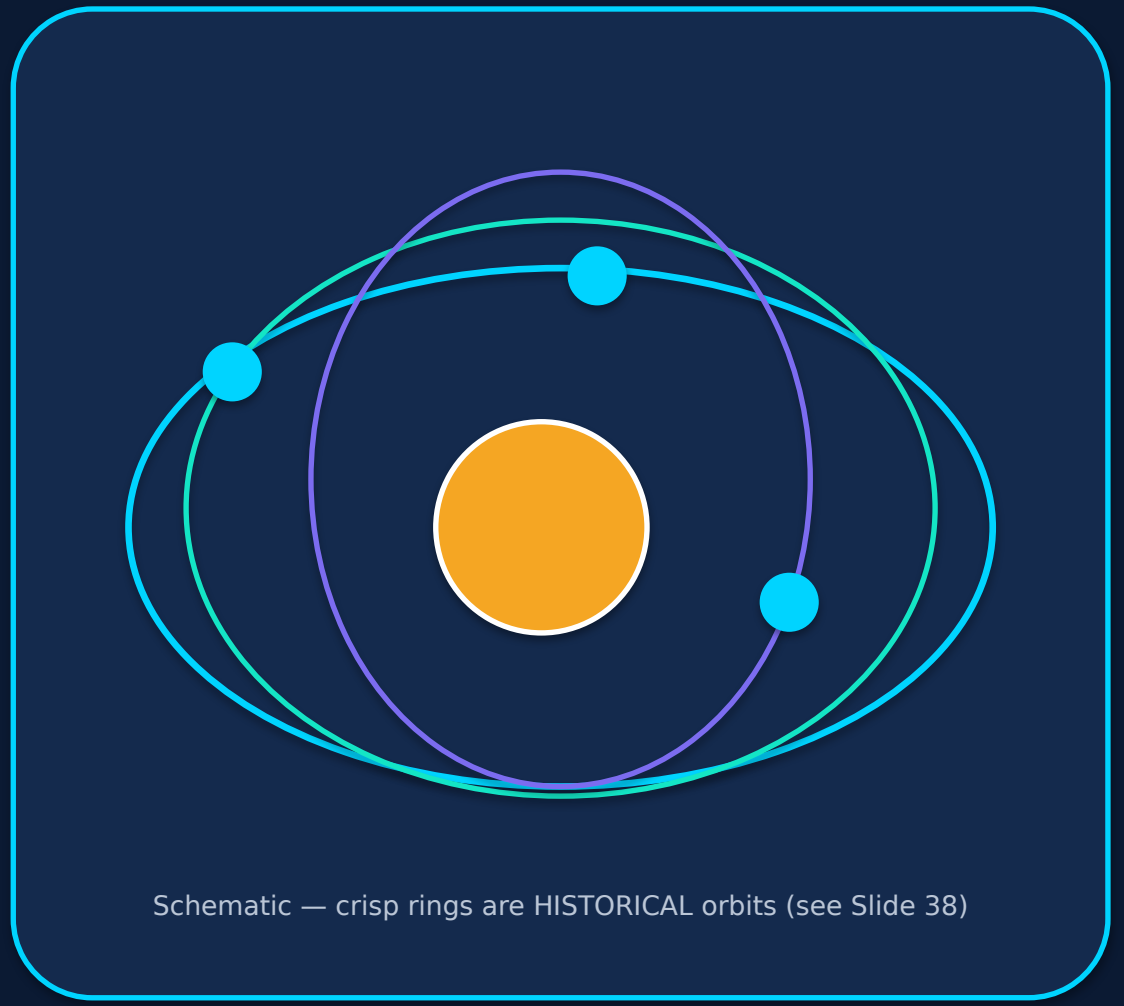
ATOMIC STRUCTURE

From Thomson's Plum Pudding Model to Modern Quantum Theory

First-year undergraduate lecture • Group of 10 • [Team Name TBD] • [College Name TBD]



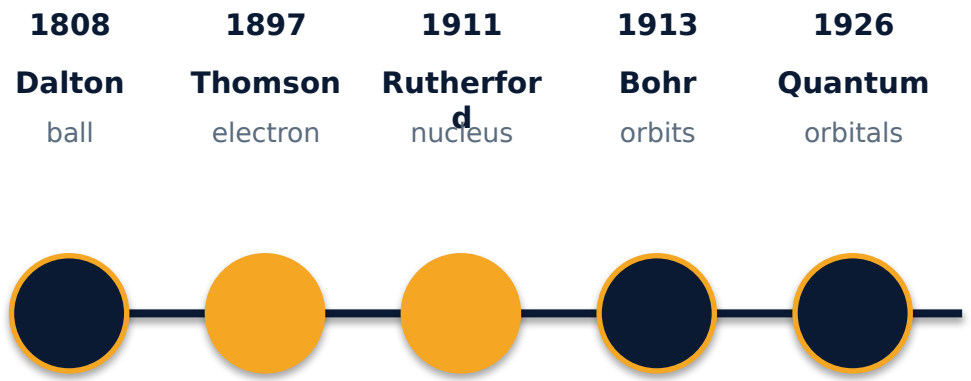
Schematic — crisp rings are HISTORICAL orbits (see Slide 38)

- Ten students teach one story: how the electron was found, how Thomson built a model, how Rutherford broke it, and why even Rutherford had to be replaced.
- Driving question for the whole hour: atoms contain charged parts, so what holds them together — and why do atoms not collapse?
- Every slide shows real dates, numbers, and equations you can verify — plus the member number responsible for teaching it.
- Team: Members 01–10 (names TBD). College: [College Name

Our Learning Journey — Five Models, One Method

ALL MEMBERS • M01

- 1808 Dalton: solid ball explains combining ratios — Member 1 shows what it explained and why it was believed.
- 1897–1904 Thomson: electron found, pudding built — Members 2–4 cover discovery, model, and its failures.
- 1909–1911 Rutherford, Geiger, Marsden: nucleus found — Members 5–7 rebuild the atom from scattering evidence.
- 1913 Bohr to quantum orbitals: stability and spectra fixed — Members 9–10 close the arc after Member 8 compares models.
- Method to carry all hour: each model kept what worked and was broken by a new experiment with numbers.

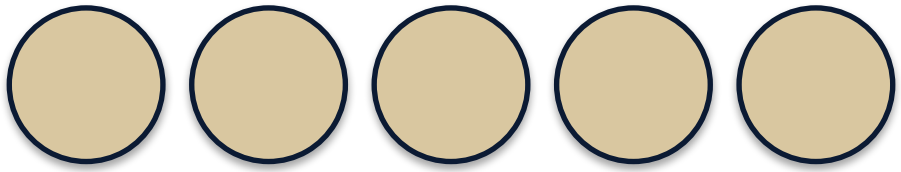


Members 01–10 own the numbered stops — follow the pills.

What Is an Atom? From Atomos to Dalton's Theory (1808)

PRESENTER 01

- The word atom comes from Greek atomos, meaning indivisible — first a philosophical idea, not an experiment.
- In 1808 John Dalton turned philosophy into testable chemistry: matter is made of tiny particles called atoms.
- Dalton's rules: atoms of one element are identical, and atoms combine in fixed ratios to form compounds.
- Example the class can weigh: carbon and oxygen always combine in fixed mass ratios, never arbitrary amounts.
- In Dalton's picture the atom was a tiny solid ball that could never be broken — a claim later experiments destroyed.



atomos — uncuttable spheres, one kind per element

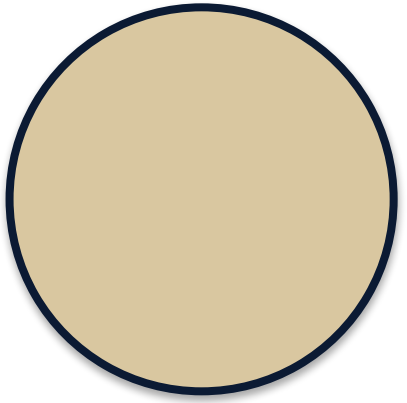


Custom schematic — fixed ratios you can weigh.

Dalton's Solid Ball — What It Explained and Why It Was Accepted

PRESENTER 01

- The solid ball explained why elements combine in fixed, predictable ratios instead of melting together arbitrarily.
- It gave chemistry measurable atoms: each element had a characteristic combining mass chemists could weigh in the lab.
- With no evidence of inner parts for nearly a century, an indivisible sphere was the simplest picture that worked.
- Acceptance came from usefulness: the model predicted combining behaviour correctly even though its hardness was assumed.
- Its weakness was structural silence: a solid ball cannot emit electrons, conduct in discharge tubes, or give line spectra.



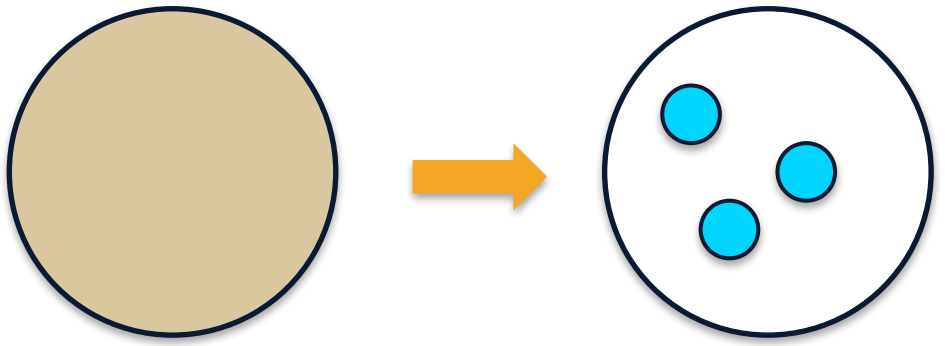
Useful: predicts combining masses • Silent: no inner parts

Kept for ~90 years because it WORKED, not because it was proven

Atoms Are Divisible — Electricity Reveals Parts Inside

PRESENTER 01

- Electrolysis splits compounds with electric current — neutral matter responds to charge, so charge must live inside atoms.
- Electric discharge through low-pressure gases glows and conducts — again, atoms and electricity are connected.
- Both experiments force one conclusion: atoms contain smaller charged parts, so atoms are divisible after all.
- This raises the question every atomic model tries to answer: how are the inner charged particles arranged?
- Atomic models are scientists' arranged-furniture plans for these parts — Member 2 opens the first drawer: the electron.



Electrolysis + discharge glow → electricity lives inside atoms

Custom schematic — the logical crowbar that opens the ball.

The Three Subatomic Particles — Electron, Proton, Neutron

PRESENTER 01

- The electron carries -1.6×10^{-19} C with a tiny mass of $\sim 9.1 \times 10^{-31}$ kg — the lightest part, first proof atoms break open.
- The proton carries the same charge but positive ($+1.6 \times 10^{-19}$ C) and is ~1836 times heavier than the electron.
- The neutron, found later by Chadwick in 1932, has no charge and a mass almost equal to the proton.
- Neutrality rule: total positive charge equals total negative charge, so the whole atom is electrically neutral.
- TABLE (right): read each row's charge, mass, and discoverer — the rest of this talk only rearranges these three parts.

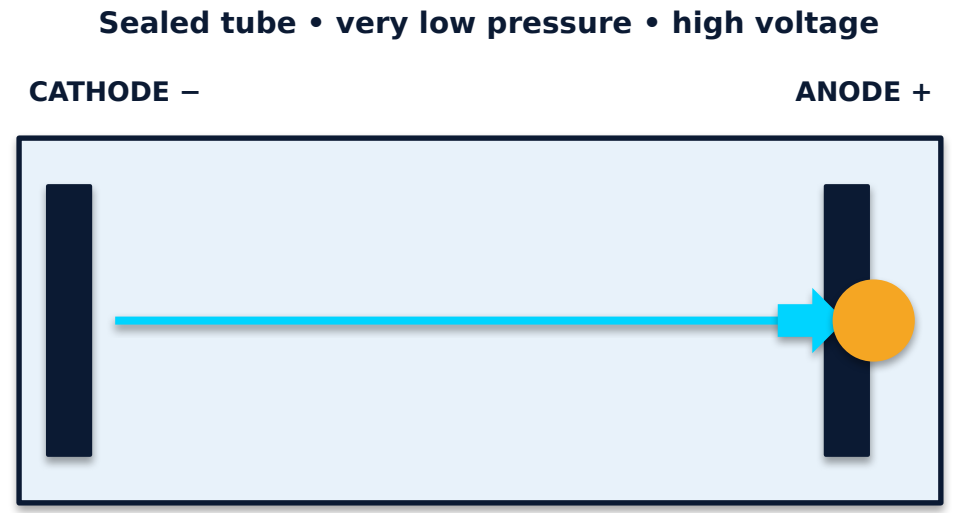
Particle	Charge	Mass	Found
electron	-1.6×10^{-19} C	9.1×10^{-31} kg	Thomson 1897
proton	$+1.6\times10^{-19}$ C	$\sim1836 \times e^-$	—
neutron	0	≈ proton	Chadwick 1932

Neutrality: total + equals total −

Discovery of the Electron (1897) — Building the Cathode-Ray Tube

PRESENTER 02

- J. J. Thomson discovered the electron in 1897 while studying cathode rays — the project starts with his glass tube.
- The tube is sealed glass holding a gas at very low pressure, so the beam flies without scattering off air molecules.
- Two electrodes sit inside: the cathode (−) and the anode (+), wired to a high-voltage supply across the tube.
- When voltage is applied, rays stream from cathode to anode — these streams are the cathode rays under test.
- Portrait: J. J. Thomson (photo, public domain) — the glow alone proved nothing; his deflection measurements proved everything.





J. J. Thomson (PD)

Straight beam = baseline for the field tests.

Custom schematic + Thomson portrait (PD).

Cathode Rays at Work — Straight Lines, Shadows, Motion, Glow

PRESENTER 02

- The rays travel in straight lines — slide any object into the beam and it casts a sharp shadow on the glass.
- Straightness plus shadow proves directed travel from cathode to anode, establishing the baseline undeflected path.
- A tiny paddle wheel placed in the beam rotates — the rays carry energy and momentum, so they are physical, not illusion.
- Where rays strike, glass or a fluorescent screen glows — each impact point becomes visible as a bright spot.
- So far we know the path and the punch, but not the charge — the field tests on the next slide decide that.



Glowing beam in a real tube (modern photo, illustrative — not Thomson's original)

- straight + shadow
- paddle spins
- screen glows

Negative Charge Proved — Bending in Electric and Magnetic Fields

PRESENTER 02

- In an electric field the rays bend toward the positively charged plate — only a negative charge is pulled that way.
- Magnetic fields deflect them likewise, confirming the beam responds as moving charge, not as light waves.
- The undeflected straight path versus the bent path is the whole argument: the difference is caused by charge.
- Light would ignore both fields; these rays obey both — so they are streams of charged particles.
- Charge is now settled; the next measurement asks how light each particle is — the famous ratio.



Bends toward + plate, so the beam is NEGATIVE. Magnetic deflection agrees.

Custom schematic — bend direction is the evidence.

How Light Is the Electron? e/m = 1.76 × 10^{11} C/kg

PRESENTER 02

- Thomson balanced electric against magnetic deflection until the two bends cancelled — then computed e/m = 1.76 × 10^{11} C/kg.
- Here e is the particle's charge in coulombs and m its mass in kilograms, so the ratio reads coulombs per kilogram.
- The value is ~1800 times larger than for a hydrogen ion — for equal charge the mass must be ~1800 times smaller.
- The same ratio appeared with every gas and every cathode metal — so electrons are a common part of every atom.
- Later Millikan's oil-drop experiment fixed the charge e separately; combined with e/m this gives the electron mass. The atom is divisible.

e/m = 1.76 × 10^{11} C/kg

E-field bends down

B-field bends up

balanced → cancel → measure

e = charge (C) • m = mass (kg) • ~1800× hydrogen ion → extremely light

Same ratio, every gas and cathode metal → electrons live in ALL atoms

Millikan oil-drop fixed e separately → mass follows from e/m

Thomson Asks the Next Question — Where Is the Positive Charge? (1904)

PRESENTER 03

- Electrons are negative, yet whole atoms are neutral — so an equal positive charge must exist somewhere inside.
- In 1904 Thomson proposed his answer: the atom is a sphere of diffuse positive charge filling the whole atom.
- The sphere's radius is ~10^{-10} metre — this size becomes the target Rutherford's experiment will test.
- Crucially, there is no nucleus in this picture: positive charge is spread smoothly everywhere, not concentrated.

e^- negative + atom neutral = missing + charge … where?



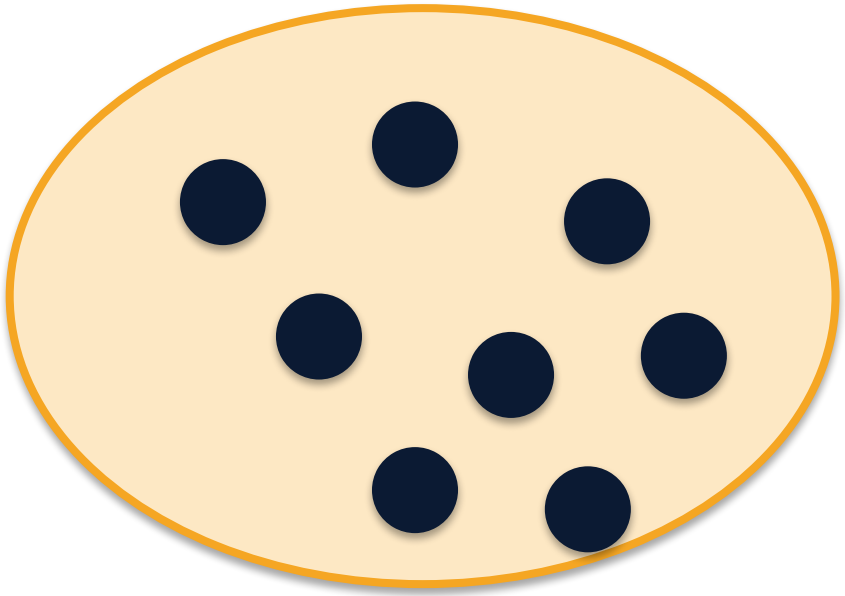
R ≈ 10^{-10} m

1904 answer: uniform + sphere, NO nucleus

PRESENTER 03

Plum Pudding — Positive Pudding, Electron Plums, Neutral Whole

- The pudding is the uniform positive charge; the plums are electrons embedded throughout it like fruit in dessert.
- Twin analogy from the source: watermelon red pulp is the positive charge and the seeds are the electrons.
- Embedding matters: electrons sit inside the positive sphere, held wherever the surrounding charge balances them.
- Neutrality is automatic: total positive charge equals total negative charge, so the atom has no net charge.
- Point at the diagram: amber fill everywhere versus navy dots — fill and dots together make neutrality visible.



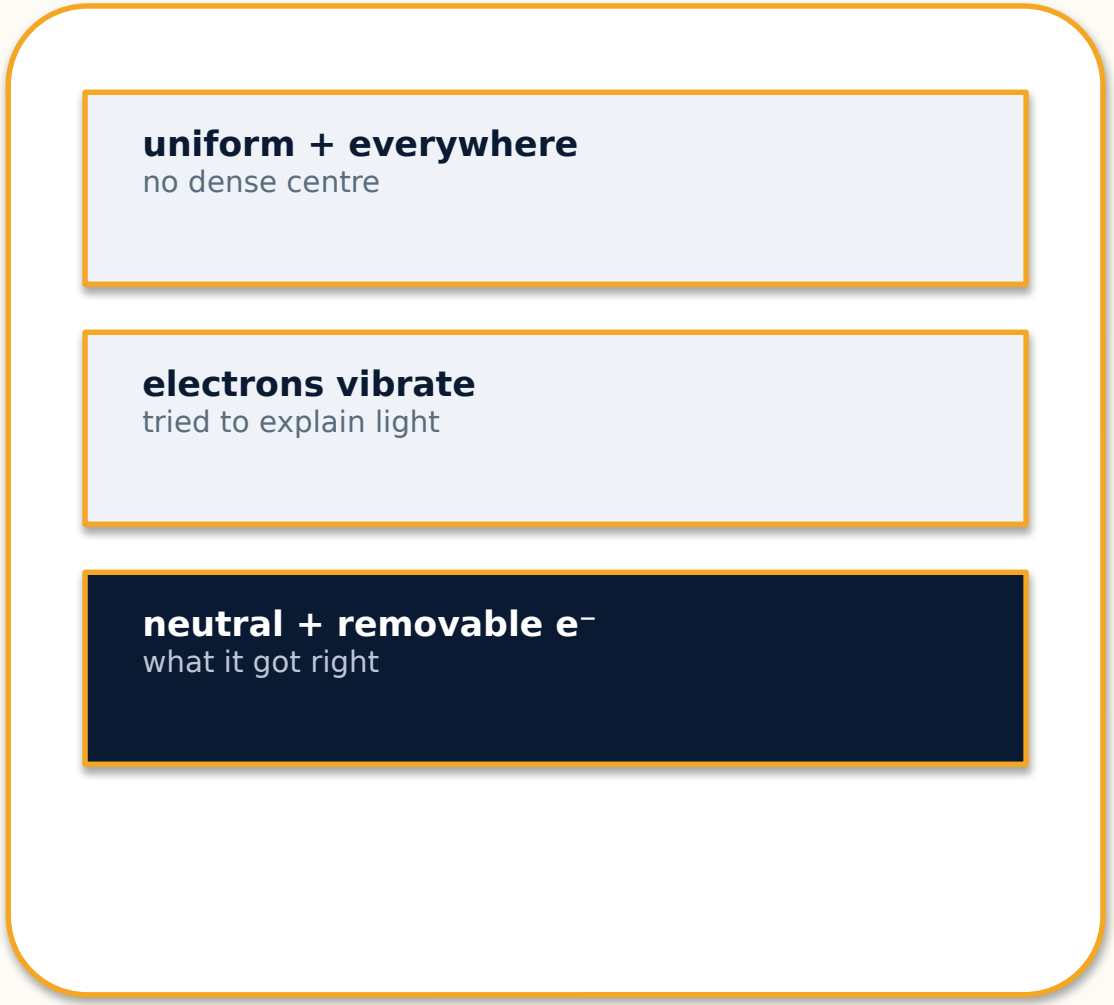
Pudding = uniform + • Plums = electrons • total + = total −

Custom schematic — watermelon twin: pulp = +, seeds = electrons.

Assumptions Inside the Pudding — And What It Could Explain

PRESENTER 03

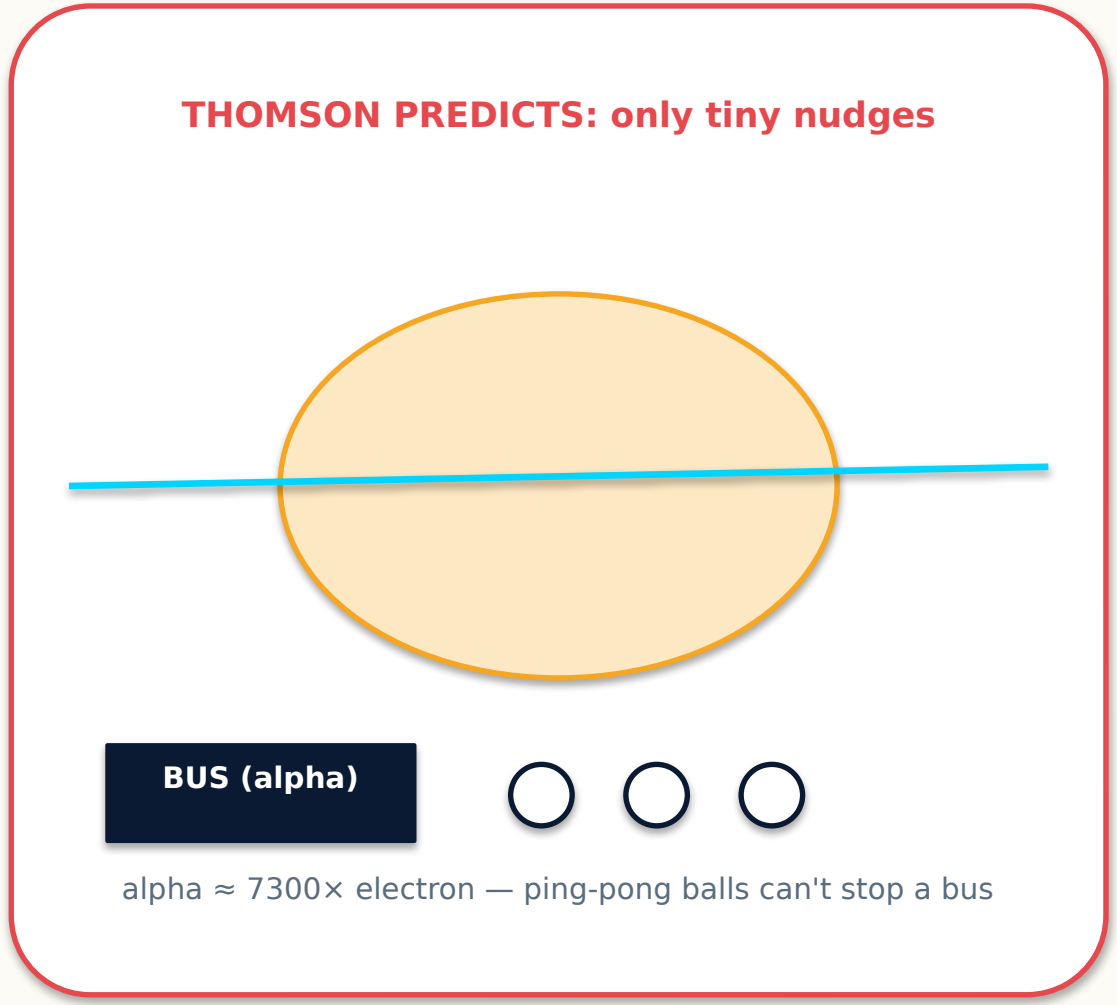
- Assumption one: positive charge is spread uniformly through the entire atomic volume, with no dense centre.
- Assumption two: electrons sit or vibrate gently about fixed positions inside that uniform charge.
- Thomson used these vibrations to try to explain light emission — shaking charges should radiate light.
- What the model genuinely explained: electrical neutrality and the fact that electrons can be pulled out of atoms.
- What it never earned: exact spectral lines, a nucleus, or stability proof — Member 4 prosecutes each gap.



Failure One — Large-Angle Scattering Is Impossible in Pudding

PRESENTER 04

- Spread-out positive charge makes only a weak electric field inside the atom, so a passing alpha feels a slight push.
- The model therefore predicts every alpha particle suffers only a tiny deflection — a small wiggle, never a rebound.
- Worse, electrons cannot rescue the prediction: an alpha is ~7300 times heavier than an electron.
- Source analogy: a bus hitting ping-pong balls — the bus never bounces back, no matter how many balls it meets.
- Experiments showed violent large-angle rebounds instead — the first direct clash between pudding and reality.



Failures Two to Four — No Nucleus, No Lines, No Stability

PRESENTER 04

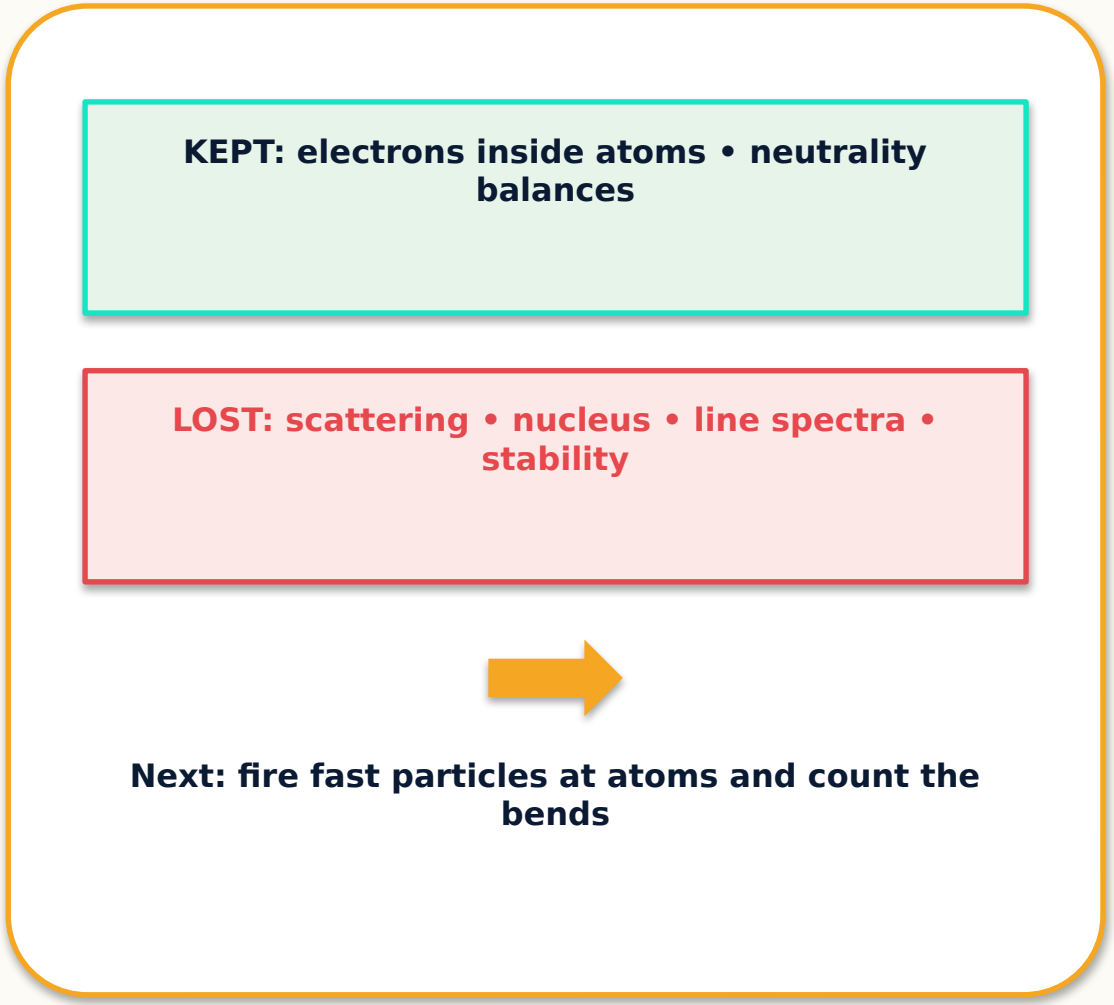
- No nucleus: the model assumes positive charge fills the atom, leaving no dense centre that could hurl an alpha back.
- No line spectra: real atoms emit light only at exact wavelengths, but vibrating electrons give no such sharp lines.
- Observe-vs-model discipline: exact wavelengths are measured facts; Thomson's smooth vibrations are the failed explanation.
- No stability: a set of motionless charges cannot sit in stable balance under electrostatic forces alone.
- Three independent failures plus scattering make four — the model is cornered on every side except neutrality.



Verdict on Thomson — And the Experiment That Will Test It

PRESENTER 04

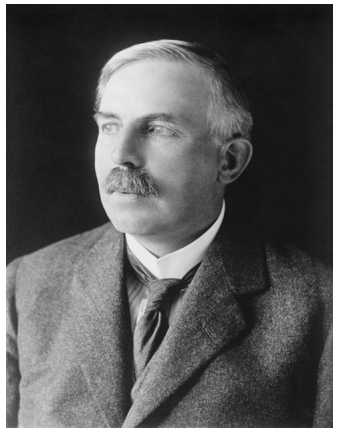
- Kept: electrons exist inside atoms, and neutrality balances equal positive against negative charge.
- Lost: large-angle scattering, nuclear structure, exact line spectra, and guaranteed stability.
- The honest scientific move is a direct test: fire fast heavy particles at atoms and watch the deflections.
- If pudding is right, almost every particle sails through with a tiny nudge — Rutherford's team ran exactly this trial.
- Transition question for the class: what apparatus could count single particles at every angle? Member 5 answers.



Testing Thomson — Rutherford, Geiger and Marsden (Manchester, ~1909)

PRESENTER 05

- Experimenters: Ernest Rutherford with Hans Geiger and Ernest Marsden, working at Manchester around 1909.
- Their aim was explicit: test Thomson's model by firing fast particles at atoms and watching how they deflect.
- Strategy: use a heavy, fast, charged probe so any concentrated charge inside the atom reveals itself by repulsion.
- Everything that follows — source, beam, foil, screen — serves that single aim: count deflections angle by angle.
- Portrait: Rutherford (photo, public domain) — Geiger and Marsden ran the bench; Rutherford read the meaning.



Rutherford • Geiger • Marsden — Manchester, ~1909

Bench: Geiger and Marsden.
Meaning: Rutherford.

AIM: test Thomson — fire fast particles, watch deflections

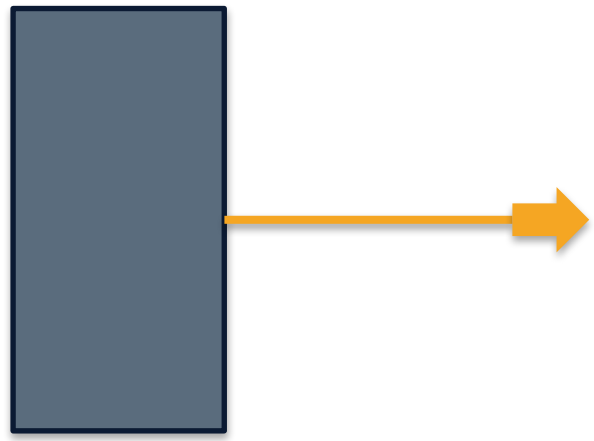
Every part below serves that aim — source, beam, foil, screen.

Portrait: Bain News Service via LoC (PD-US), restoration Bammesk, via Commons.

The Probe — Alpha Particles in a Lead Collimator

PRESENTER 05

- The source is a radioactive material such as radium or polonium, steadily emitting alpha particles.
- Alpha particles are helium nuclei — two protons plus two neutrons — carrying charge +2e.
- They leave the source at ~10^7 metres per second: heavy and fast, ideal hammers for testing atomic structure.
- The source sits in a lead block with a narrow hole; lead absorbs every stray particle except the thin emerging beam.
- Result: a narrow, aimed, single-direction beam — without collimation the angle counts would be meaningless.



Pb block + hole

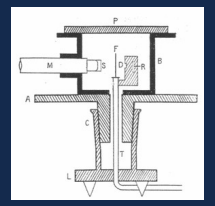
alpha = helium nucleus (2p + 2n) • +2e • ~10^7 m/s

radium / polonium source • lead absorbs the rest → thin aimed beam

PRESENTER 05

Target and Detector — Gold Foil, Vacuum, Flashes Counted One by One

- The beam strikes a gold foil only a few hundred atoms thick — nearly two-dimensional, so each alpha meets few atoms.
- Gold was chosen for two reasons: it hammers into extremely thin sheets, and its heavy nucleus (charge +79) repels strongly.
- Around the foil stands a zinc-sulphide screen: every alpha impact flashes as a tiny scintillation of light.
- A microscope counts these flashes angle by angle — single-particle detection, position by position, in the dark.
- The whole apparatus sits in vacuum so alphas never scatter off air; Thomson's expectation was almost-all-through with tiny bends.





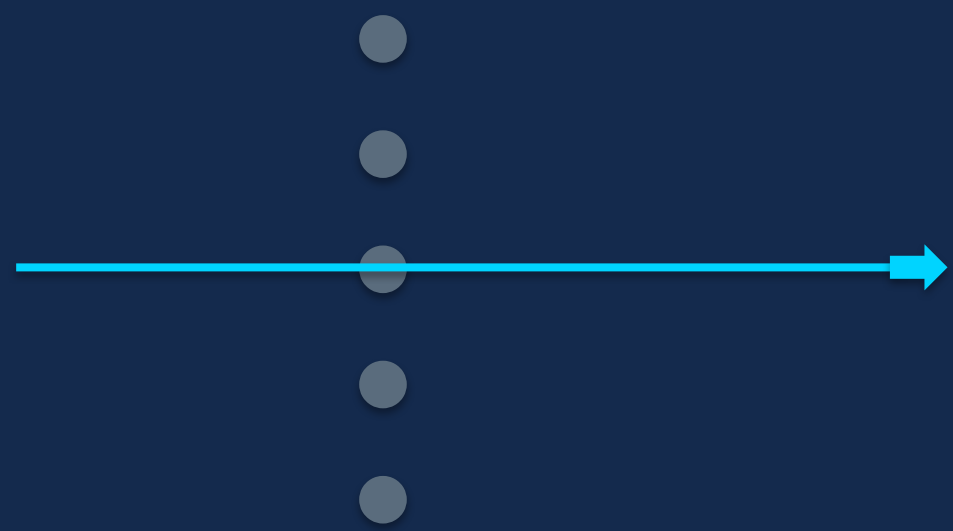
few-hundred atoms thick • malleable • vacuum • microscope counts

Custom top-view schematic + historic Geiger–Marsden fig. thumbnail (PD, 1913).

Observation 1 — Almost Everything Goes Straight Through

PRESENTER 06

- What happened: the overwhelming majority of alpha particles crossed the foil with zero measurable deflection.
- This is a seen count, not an inference — flash after flash appeared at the straight-ahead position.
- Conclusion, and only this: most of the atom offers nothing solid enough to disturb a fast heavy particle.
- In the source's words: the atom is mostly empty space — the beam usually meets no charge worth noticing.
- Keep observation and conclusion apart: straight flashes were seen; emptiness is what they mean. Two more results follow.

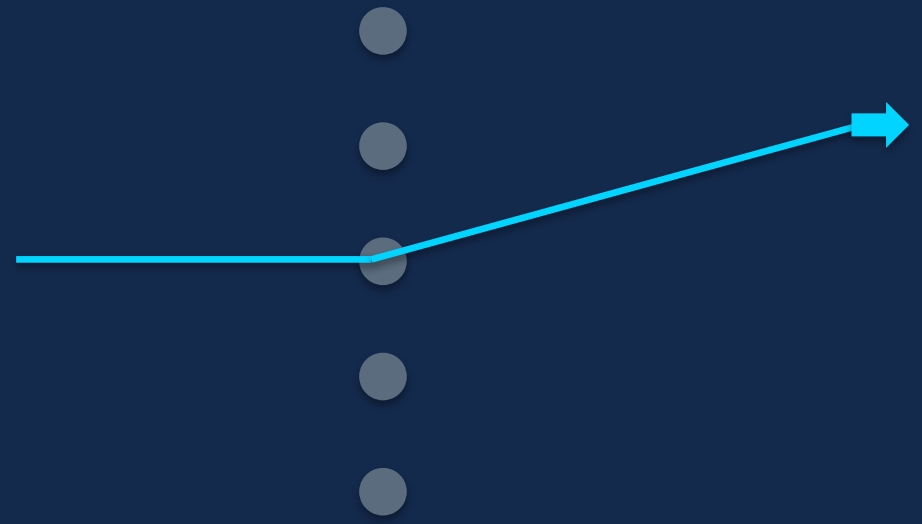


Custom schematic — left column is SEEN, right column is MEANT.

Observation 2 — A Few Particles Bend by Small Angles

PRESENTER 06

- What happened: a small fraction of alphas left the foil travelling a few degrees off the straight path.
- Source figure: about 0.14% of particles deflected by more than 1 degree — rare, but far too many to ignore.
- Conclusion: positive charge really exists inside the atom, and it repelled the positively charged alphas.
- Like charges repel: the bend direction and size fingerprint a positive interior acting at a distance.
- These are near misses past buried charge — the next result shows what a direct hit looks like.

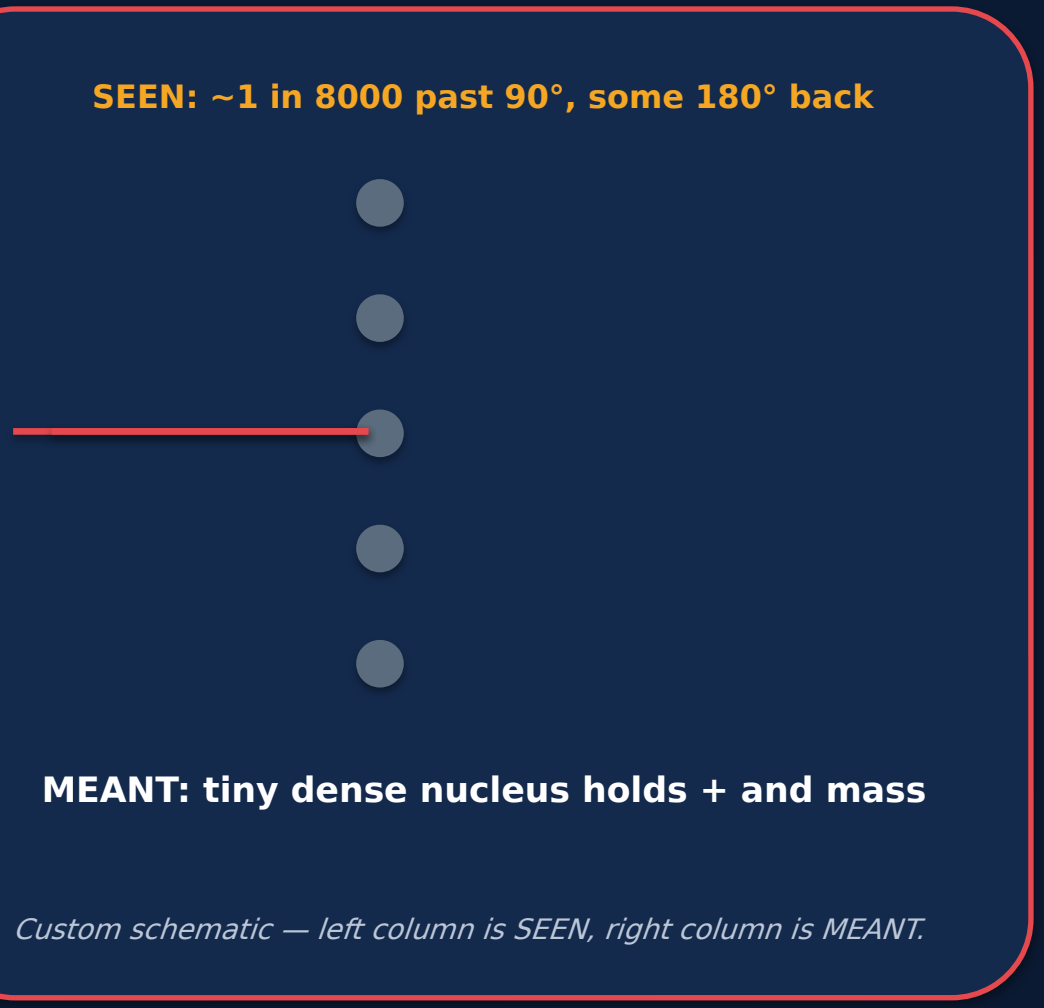


Custom schematic — left column is SEEN, right column is MEANT.

Observation 3 — About 1 in 8000 Bounces Back

PRESENTER 06

- What happened: very few alphas — about 1 in 8000 — scattered through more than 90 degrees, some straight back at 180°.
- Conclusion: almost all the atom's mass and all its positive charge sit in a tiny, dense nucleus.
- Reasoning: only a head-on collision with concentrated charge can reverse a fast, heavy alpha particle.
- Rutherford's image: like firing a huge shell at tissue paper and watching it come back — utterly unexpected.
- This single result kills spread charge: pudding predicts zero rebounds, yet rebounds were counted flash by flash.

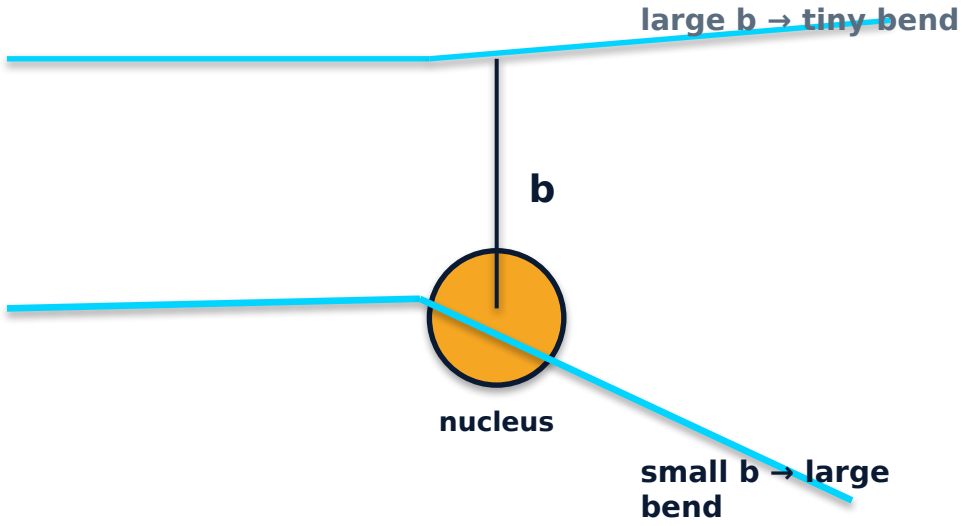


Custom schematic — left column is SEEN, right column is MEANT.

Impact Parameter — Why the Same Beam Gives Three Fates

PRESENTER 06

- Definition: the impact parameter b is the perpendicular distance between the alpha's original path and the nucleus centre.
- In plain words, b measures how off-centre the aim is — a wide miss has large b, a near hit has small b.
- Large b: the alpha passes far away, feels only weak distant repulsion, and sails on nearly straight — the common fate.
- Small b: the alpha dives through the strong near-nucleus field and swings through a large angle — the rare 0.14% tail.
- Diagram: trace both paths from the same incoming direction; the nucleus sits at the bend centre of each.

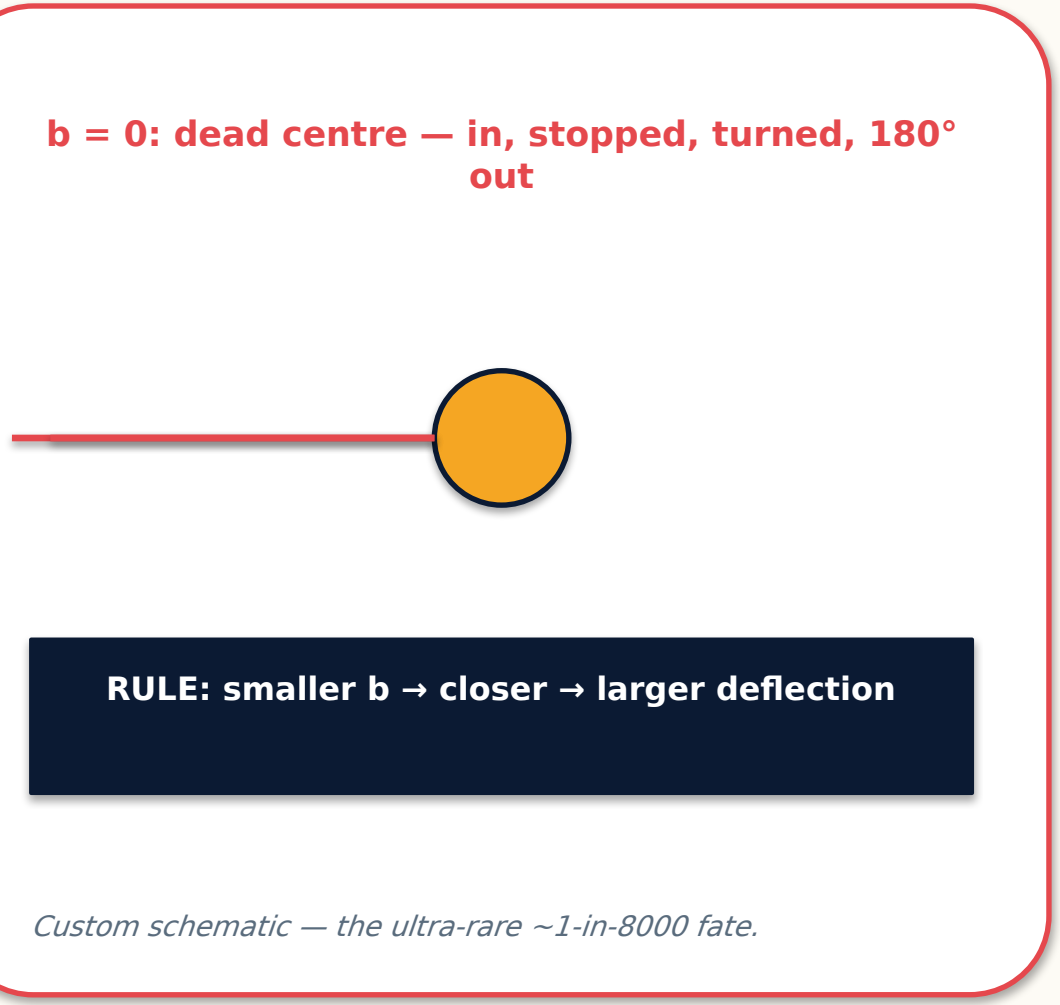


Custom schematic — b is measured from the original path to the centre.

Head-On Collision — When b = 0 the Particle Comes Back

PRESENTER 06

- Zero impact parameter means perfectly head-on aim: the alpha charges straight at the nucleus centre.
- It climbs the repulsion head-on, stops, turns around, and exits back along its entry line at 180 degrees.
- This is the ultra-rare ~1-in-8000 fate — direct hits are rare because nuclei are vanishingly small targets.
- One-line rule to memorize: smaller impact parameter means closer approach means larger deflection.
- From this data Rutherford built a new model — Member 7 assembles it from exactly these three fates.

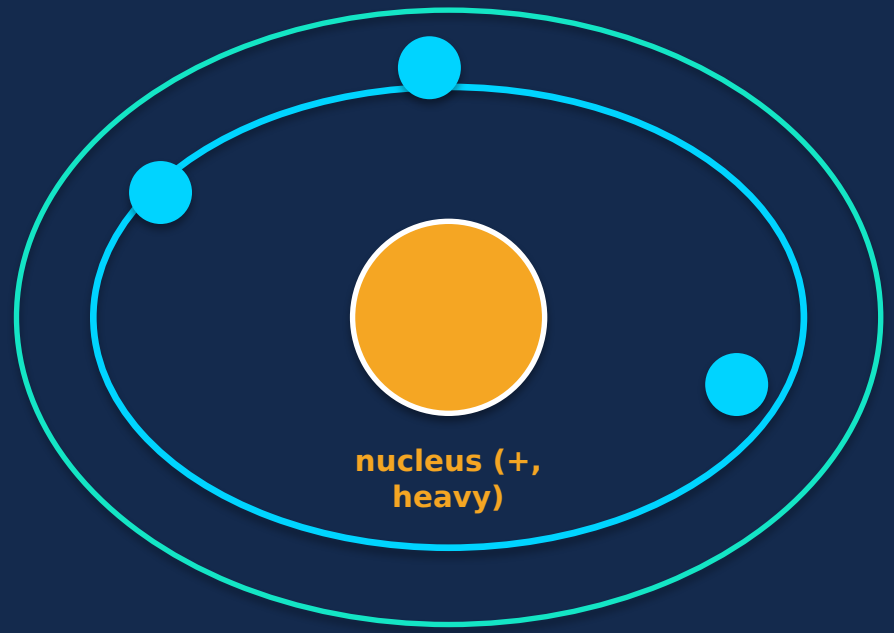


Custom schematic — the ultra-rare ~1-in-8000 fate.

Rutherford's Nuclear Model (1911) — A Dense Centre in Vast Emptiness

PRESENTER 07

- In 1911 Rutherford proposed: a small, dense, positively charged nucleus sits at the atom's centre.
- Most of the atom's mass lives in that nucleus — the light electrons contribute almost nothing to the weight.
- Electrons revolve around the nucleus in circular orbits, and the rest of the atom is empty space.
- Because electrons circle like planets around the Sun, this is also called the planetary model.
- Historical label: the orbit picture below explains scattering beautifully — but it is the old model, not today's quantum atom.



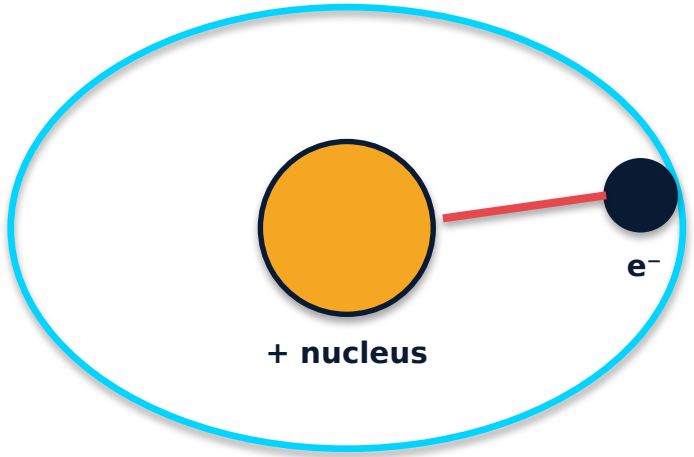
HISTORICAL orbit picture — kept for scattering, replaced for stability

What Holds Electrons Up? Coulomb Attraction as Centripetal Force

PRESENTER 07

- Without something pulling inward, orbiting electrons would fly off in straight lines — circular motion needs a centre pull.
- That pull is electrostatic attraction between the positive nucleus and each negative electron.
- Just as gravity holds planets, charge holds electrons — with attraction playing gravity's role in the analogy.
- Historical relation: F(Coulomb) = m·v²/r — the inward electric force supplies the required circular-motion force.
- Term by term: F is the Coulomb attraction in newtons, m the electron mass, v its orbital speed, r the orbit radius.

F(Coulomb) = m·v²/r



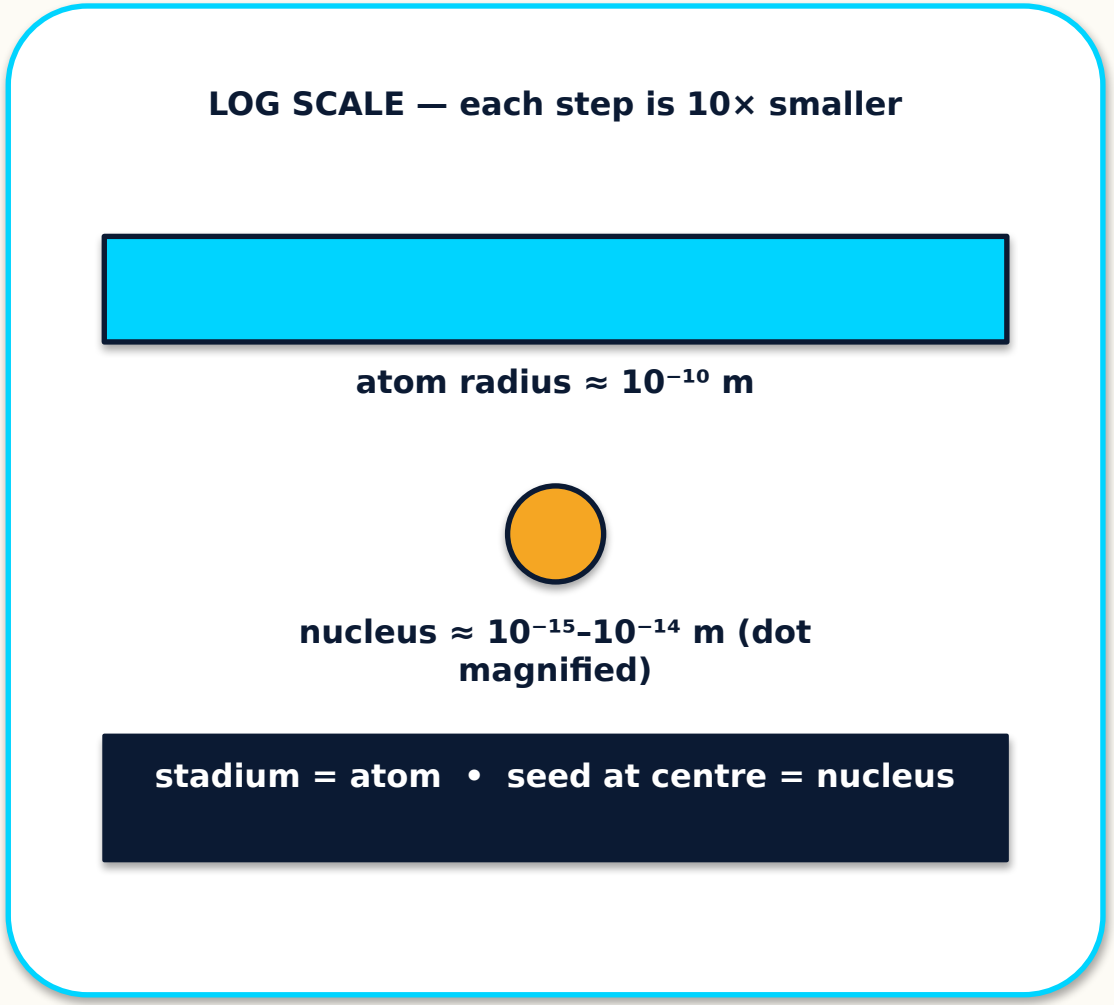
F attraction (N) • m electron mass • v speed • r radius

Custom schematic — red arrow is the inward pull; gravity plays this role for planets.

How Small Is the Nucleus? The Stadium-and-Seed Scale

PRESENTER 07

- Source sizes: the nucleus radius is ~10^{-15} to 10^{-14} m, while the whole atom's radius is ~10^{-10} m.
- So the atom is roughly 10,000 to 100,000 times wider than its nucleus — emptiness dominates by four to five orders.
- Source analogy: if the atom were a football stadium, the nucleus would be a tiny seed at the centre spot.
- The log bar (right) is honest about drawing: both sizes cannot share one linear scale, so the dot is magnified.
- Memorize both powers of ten — the teacher's favourite trap is asking which belongs to nucleus and which to atom.



What the Nuclear Model Wins — And What It Still Owes

PRESENTER 07

- It explains straight-through flight: most alphas never approach any nucleus, so most flashes sit dead ahead.
- It explains small bends: distant passes feel weak repulsion — the common 0.14% tier falls out naturally.
- It explains rebounds: the nucleus packs strong field into a tiny volume, so rare direct hits bounce back.
- One concentrated centre replaces pudding's smear — every observation tier now has a matching mechanism.
- What it still owes: why orbiting electrons neither radiate nor spiral in, and why spectra are sharp lines — Member 9 collects.

straight flight
most alphas never near a nucleus

small bends
distant passes feel weak push

rare rebounds
direct hits meet dense + core

Still owed: silent stable orbits + sharp spectral lines

One concentrated centre explains all three tiers.

Thomson vs Rutherford — Same Electrons, Opposite Architecture

PRESENTER 08

- One-line contrast: Thomson imagined a uniform FILLED atom; Rutherford proved a mostly EMPTY atom with a tiny dense centre.
- TABLE (right): positive charge spread through the atom versus concentrated in the nucleus — the whole difference.
- Electrons embedded in the sphere versus revolving around the nucleus; no nucleus versus a dense massive one present.
- Alpha scattering: Thomson predicts only tiny bends and fails, while Rutherford predicts all three tiers and succeeds.
- Bridge warning: both pictures are still classical — and classical orbits carry the fatal flaw Member 9 now exposes.

Feature	Thomson	Rutherford
Positive charge	Spread through atom	In tiny nucleus
Electrons	Embedded in sphere	Revolve in orbits
Nucleus	None	Dense, massive
Alpha scattering	Fails: tiny bends only	Explains all 3 tiers

Both classical — the orbit flaw falls next.

Five Models in One View — Structure, Evidence, and What Broke Each

PRESENTER 08

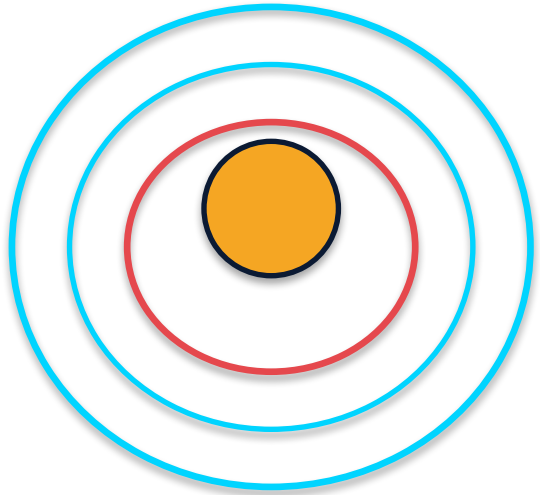
- Dalton's ball: solid and indivisible — explained combining ratios until discharge tubes proved inner charge.
- Thomson's pudding: electrons in uniform positive — explained neutrality until large-angle scattering broke it.
- Rutherford's nucleus: dense centre with orbiting electrons — explained scattering until collapse and spectra broke it.
- Bohr's orbits: quantised rings giving hydrogen lines — explained spectra until heavier atoms overflowed it.
- Quantum clouds: probability orbitals as the current model — every step kept what worked and fixed what failed.

Model	Structure	Explained	Broken by
Dalton 1808	Solid ball	Combining ratios	Discharge charge
Thomson 1904	e^- in uniform +	Neutrality	Big scattering
Rutherford 1911	Nucleus + orbits	Scattering tiers	Collapse + lines
Bohr 1913	Quantised rings	H line spectrum	Heavier atoms
Quantum 1926	Probability clouds	Stability + spectra	Current model

Why Classical Orbits Collapse — The ~10^{-11} s Death Sentence

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- Circular motion constantly changes direction, and changing direction is acceleration — orbiting means accelerating.
- Classical electrodynamics (Maxwell) says an accelerated charge radiates electromagnetic waves and loses energy.
- Losing energy shrinks the orbit, which speeds the spiral: the electron should crash into the nucleus.
- The source's timescale is ~10^{-11} seconds — clearly labelled a CLASSICAL PREDICTION, never a measurement.
- Reality check: real atoms are stable indefinitely — so the prediction refutes the classical orbit, not the atom.



turning = accelerating → radiates → loses energy → spirals in

collapse in ~10^{-11} s — a PREDICTION, never measured

Continuous Glow Predicted — Sharp Lines Observed

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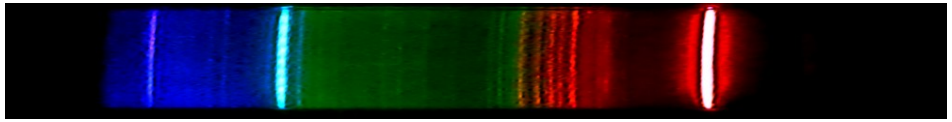
- A spiralling electron's circling frequency would change smoothly, so classical physics predicts a smooth rainbow continuum.
- Continuous means every wavelength present — a smeared band with no gaps, the signature of unquantised motion.
- Real hydrogen instead shows sharp isolated lines — the visible Balmer series in red, cyan-blue, and violet (photo, CC0).
- Those line positions follow the Rydberg pattern [SUPPL: $1/\lambda = R(1/n_1^2 - 1/n_2^2)$, $R \approx 1.097\times10^7\ m^{-1}$] — exact integers in, exact lines out.
- A spiral can never print integers: Rutherford's model is silent where hydrogen speaks most precisely.

PREDICTED: smooth continuum



MEASURED: sharp Balmer lines

(photo: hydrogen discharge, Umop503, CC0)



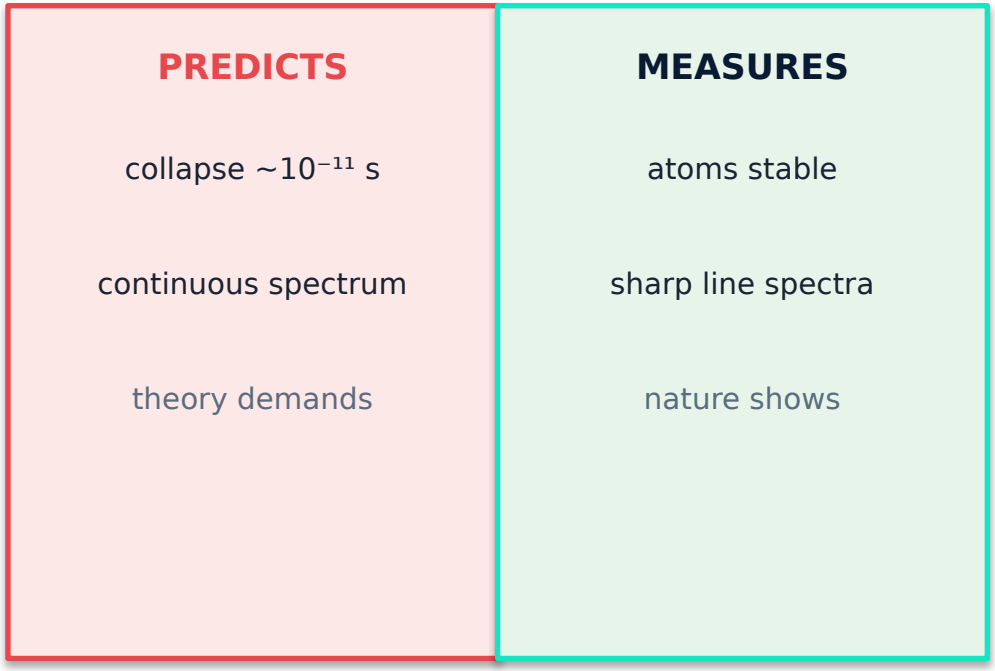
Rydberg pattern [SUPPL]: $1/\lambda = R(1/n_1^2 - 1/n_2^2)$, $R \approx 1.097\times10^7\ m^{-1}$

Integers in, exact lines out — spirals cannot print integers.

PRESENTER 09

Prediction vs Measurement — How Physics Judges a Theory

- Classical PREDICTIONS: atomic collapse in ~10^{-11} s and a continuous spectrum — what the old theory demands.
- Experimental MEASUREMENTS: atoms are stable and spectra are sharp lines — what nature actually shows.
- Physics keeps measurements and discards the theory: classical laws fail at the atomic scale, however grand elsewhere.
- This is not a draw — two independent failures (stability, spectra) convict the classical orbit twice over.
- A replacement must quantise something — Member 10 shows Bohr's first working fix.



Physics keeps measurements. Classical law fails at atomic scale.

Bohr's Fix, Idea 1 (1913) — Stationary Orbits That Do Not Radiate

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- In 1913 Niels Bohr repaired the collapse by decree: electrons may occupy certain stationary orbits.
- Inside a stationary orbit the electron radiates no energy at all — stability is imposed as a rule, not derived.
- This directly cancels the Maxwell death sentence: no radiation means no energy loss and no spiral.
- The price is honesty: Bohr cannot yet say WHY these orbits are silent — he postulates it and checks the payoff.
- Open question left hanging: which orbits qualify? Idea 2 answers with whole numbers.





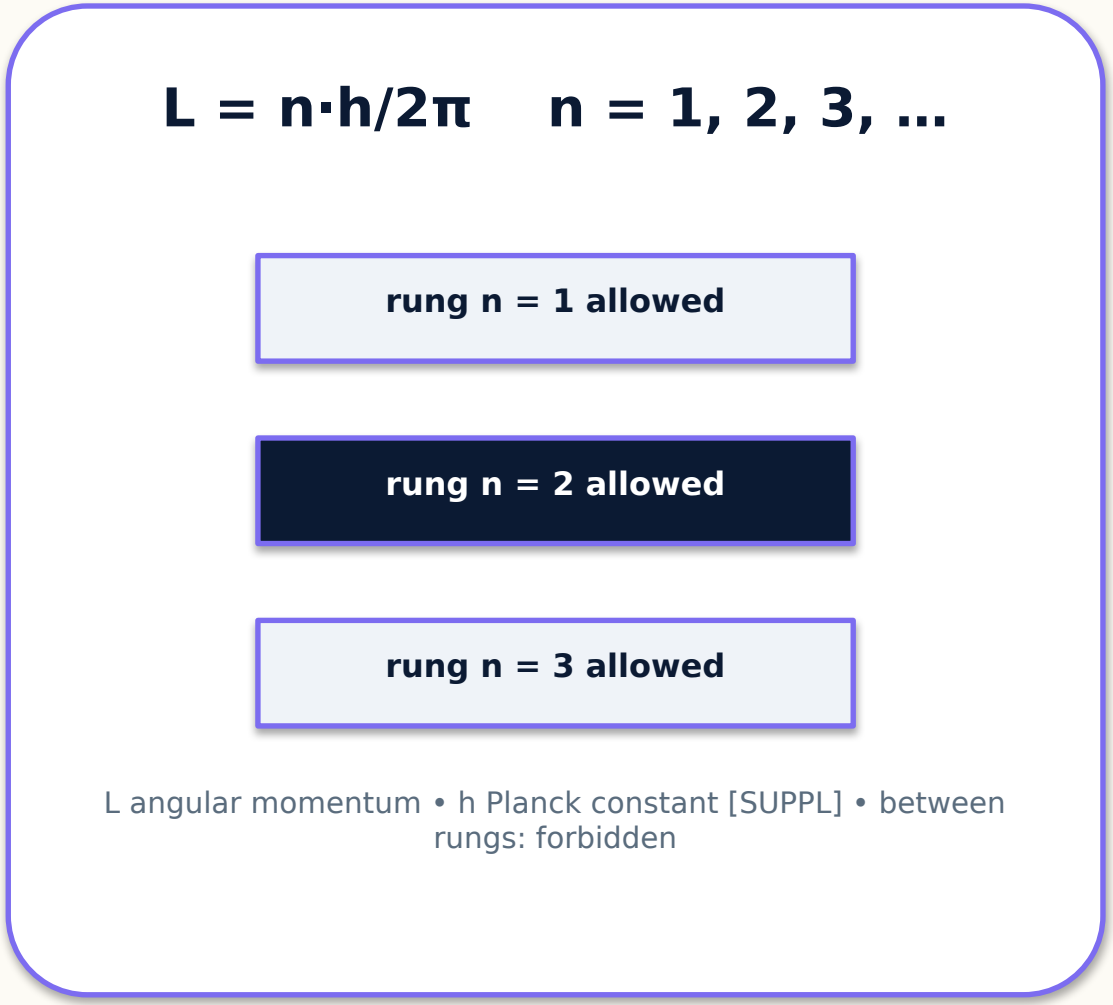
stationary ring: silent cruise, no radiation

Bohr c.1922 (PD, Nobel year) — stability postulated, then tested

Bohr's Fix, Idea 2 — Only Whole-Number Orbits Are Allowed

PRESENTER 10

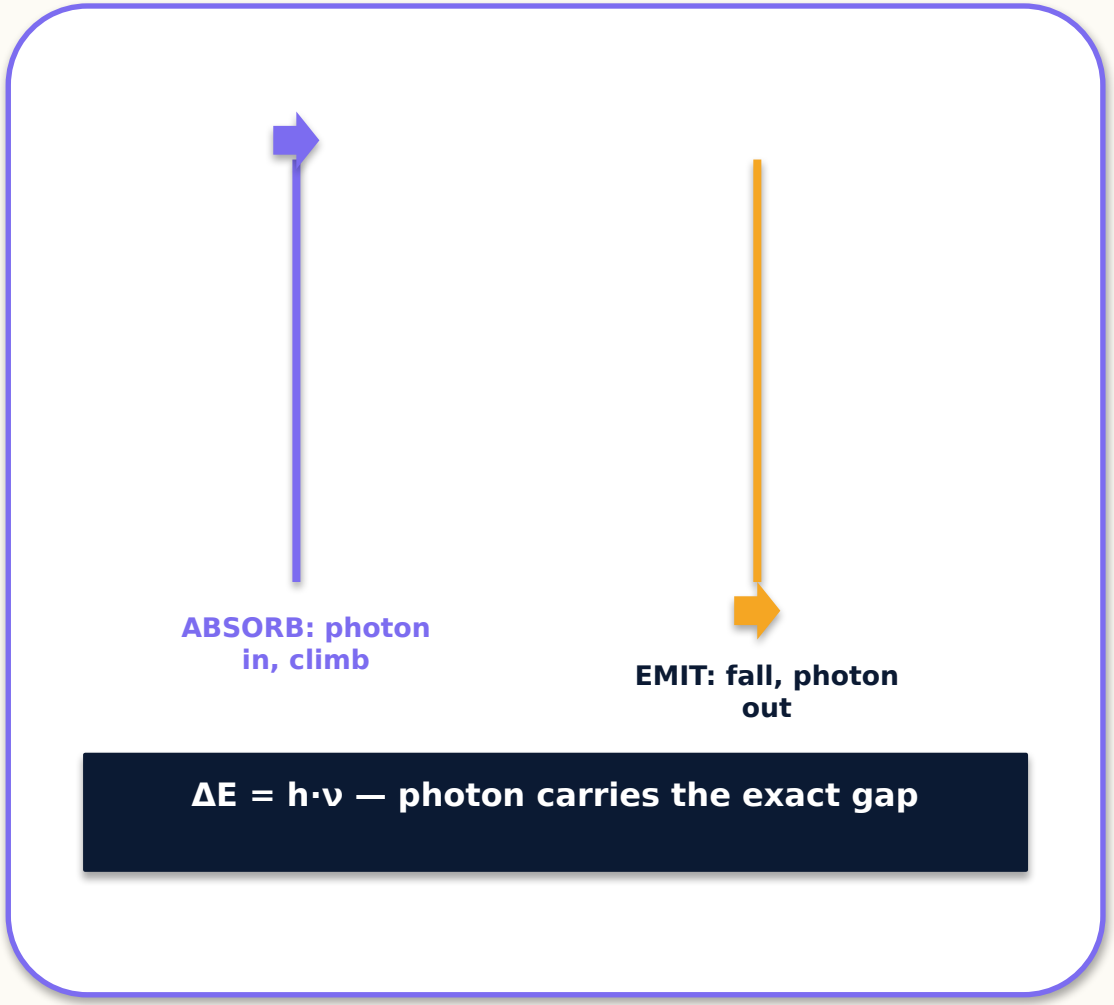
- Allowed orbits satisfy L = n·h/2π, with n = 1, 2, 3, and so on — no fractions, no in-betweens.
- Term by term: L is the electron's angular momentum, n the orbit number, h Planck's constant [SUPPL: h = 6.626×10^{-34} J·s].
- Physically this says circling motion comes in whole steps — orbits are quantised like rungs, not a ramp.
- Forbidden zones follow: an electron can sit on rung 1 or 2, but never hover halfway between them.
- With silent rungs fixed, one rule remains: how does light ever get in or out? Idea 3 answers.



Bohr's Fix, Idea 3 — Light Only in Quantum Jumps

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- An electron emits or absorbs light ONLY when jumping between allowed orbits — never while cruising inside one.
- Each jump's photon carries exactly the level difference: ΔE = h·ν, with ν the light's frequency.
- Bigger jump means bluer, more energetic light; smaller jump means redder light — colour encodes the gap.
- Fixed gaps give fixed photon energies, hence fixed spectral lines — hydrogen's spectrum is explained at last.
- The ladder and its arithmetic come next: numbers in, wavelengths out.

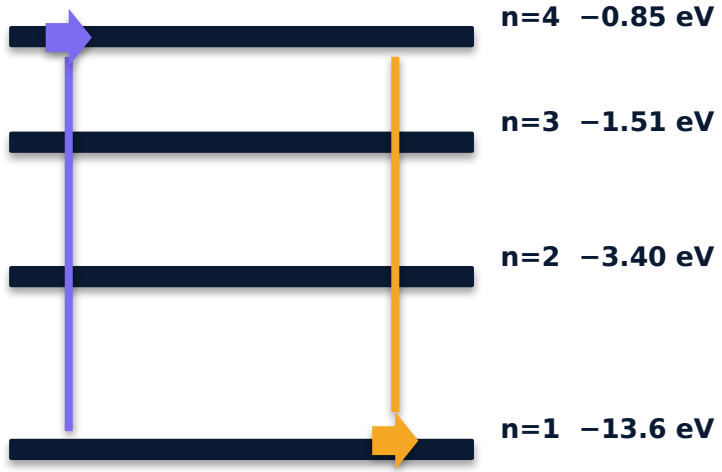


Hydrogen's Energy Ladder — E(n) = −13.6/n² eV, Worked Live

PRESENTER 10

- Source formula, hydrogen only: E(n) = −13.6/n² eV — n = 1 gives −13.6, n = 2 gives −3.4, n = 3 gives −1.51, n = 4 gives −0.85.
- UP arrow means absorption: a photon lifts the electron to a higher rung; DOWN arrow means emission as it falls.
- Worked fall 2→1: −3.4 minus −13.6 releases 10.2 eV — an ultraviolet photon of the Lyman series.
- Worked fall 3→2: −1.51 minus −3.4 releases 1.89 eV — about 656 nm red light, Balmer-α [SUPPL: hc ≈ 1240 eV·nm].
- Scope warning from the source: Bohr fits hydrogen-like atoms only — heavier atoms need the full quantum machinery.

E(n) = −13.6/n² eV (hydrogen only)

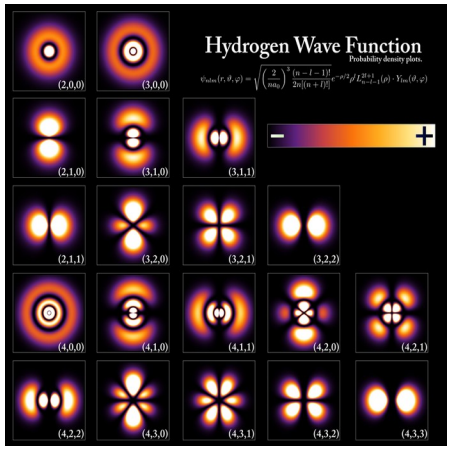


2→1: 10.2 eV UV • 3→2: 1.89 eV ≈ 656 nm red [SUPPL hc]

Beyond Bohr — Probability Clouds Replace Fixed Paths

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- Heavier atoms overflow Bohr's single-electron arithmetic, so physics rebuilt the foundation as wave mechanics.
- Heisenberg's uncertainty principle [SUPPL, 1927]: an electron's exact path and speed cannot be known together.
- Schrödinger's wave equation [SUPPL, 1926] replaces paths with wavefunctions, where $|\psi|^2$ is the probability per region.
- An orbital is such a region — the 1s cloud is densest at the centre (image: brighter means likelier; public domain).
- Exam trap, say it twice: a Bohr ORBIT is a fixed ring, a quantum ORBITAL is a probable cloud — never swap the words.
- Orbitals carry quantum numbers n, l, m, s — the address system behind the periodic table you already use.



brighter = likelier (PoorLeno, PD)

ORBIT: fixed ring (Bohr)

ORBITAL: probable cloud (now)

$|\psi|^2$ = probability

numbers n, l, m, s

PRESENTER 10

Evolution Timeline — Dalton to Quantum in Five Breaks

- 1808 Dalton's ball explained ratios — broken by discharge tubes proving inner charge.
- 1897–1904 Thomson's electron and pudding explained neutrality — broken by 1-in-8000 rebounds.
- 1909–1911 Rutherford's nucleus explained scattering — broken by predicted collapse and continuum.
- 1913 Bohr's quantised orbits explained hydrogen's lines — limited to hydrogen-like atoms.
- 1926 quantum orbitals hold today: each model kept what worked, and every break came with numbers, not opinions.



Each break came with numbers — experiment corrects theory.

Quick Revision A — Electron, Thomson, Gold Foil, Nucleus

ALL MEMBERS

- Electron? Thomson, 1897, cathode rays, e/m = 1.76×10^{11} C/kg — same ratio for every gas and cathode metal.
- Thomson model? Positive sphere ~10^{-10} m, embedded electrons, neutral overall, NO nucleus — pudding or watermelon.
- Gold foil? 1909, +2e alphas at ~10^{7} m/s onto thin gold (Z = 79), flashes on ZnS counted in vacuum.
- Three results? Most straight (empty space), ~0.14% small bends (positive charge), ~1 in 8000 back (tiny dense nucleus).

1. **1897 + e/m?**
2. **pudding = ?**
3. **gold foil = ?**
4. **3 results?**

Answers are the slide-left points — one breath each.

Quick Revision B — Collapse, Spectra, Bohr, Numbers Drill

ALL MEMBERS

- Collapse? Orbiting charge should radiate and spiral in ~10^{-11} s (PREDICTION) — yet atoms are stable.
- Spectra? Classical predicts continuum; hydrogen shows Balmer lines following the Rydberg pattern.
- Bohr? Stationary orbits, $L = n \cdot h/2\pi$, jumps with $\Delta E = h\nu$, hydrogen levels $-13.6/n^2$ eV — orbitals replace orbits today.
- Numbers drill: atom 10^{-10} m, nucleus 10^{-15}–10^{-14} m, alpha 7300× electron, proton 1836× electron, stadium holds a seed.

1. **collapse?**
2. **spectra?**
3. **Bohr?**
4. **drill numbers?**

Answers are the slide-left points — one breath each.

Teacher Cross-Questions 1–5 — Answered by the Topic Owner

MEMBERS — SEE TAGS

- Q1 [MEMBER 01]: What is an atom? The smallest particle of an element in chemical reactions — once called indivisible, now known to hold electrons, protons, neutrons.
- Q2 [MEMBER 03]: Why plum pudding? Electrons embedded in a positive sphere, like plums in pudding — or seeds in watermelon pulp.
- Q3 [MEMBER 02]: What are cathode rays? Streams of negative particles, electrons, flying cathode to anode in a low-pressure tube, bent by fields.
- Q4 [MEMBER 04]: Why did Thomson fail? No nucleus, no large-angle scattering, no exact line spectra, no guaranteed stability.
- Q5 [MEMBER 05]: What are alpha particles? Helium nuclei — two protons plus two neutrons — charge +2e, ~10^7 m/s, from radium or polonium.

Rule: the tagged owner stands, answers in two sentences, sits.

M01 answers Q1

M03 answers Q2

M02 answers Q3

M04 answers Q4

M05 answers Q5

Full Q + A pairs are on the slide left.

Teacher Cross-Questions 6–10 — Answered by the Topic Owner

MEMBERS — SEE TAGS

- Q6 [MEMBER 06]: Why did most alphas pass straight? The atom is mostly empty space — most never came near any nucleus.
- Q7 [MEMBER 06]: Why were a few greatly deflected? They passed very close to the tiny dense positive nucleus; small b means large bend, b = 0 means 180° back.
- Q8 [MEMBER 07]: What is the nuclear model? A small dense positive nucleus holds most mass, electrons circle it, rest is empty space.
- Q9 [MEMBER 07]: Why planetary? Electrons circle the nucleus like planets circle the Sun, with charge attraction replacing gravity.
- Q10 [MEMBER 09]: Classical limits? Orbits should radiate, collapse in ~10^{-11} s, and glow continuously — real atoms are stable with sharp lines.

Rule: the tagged owner stands, answers in two sentences, sits.

M06 answers Q6

M06 answers Q7

M07 answers Q8

M07 answers Q9

M09 answers Q10

Full Q + A pairs are on the slide left.

Conclusion — From Solid Ball to Probability Cloud

MEMBERS 01–10

- Dalton (1808) gave us atoms as solid balls — broken open by electricity into electrons, protons, neutrons.
- Thomson (1897/1904) gave us the electron and the first inner model — neutral but nucleus-free, broken by scattering.
- Rutherford (1909–1911) gave us the nucleus and the planetary picture — vast emptiness around a dense seed.
- Bohr (1913) quantised the orbits and saved hydrogen's lines; quantum theory replaced orbits with orbitals.
- Respected teacher and friends, thank you for listening to our presentation. We are ready for questions.

BALL → PUDDING → NUCLEUS → ORBITS → CLOUDS

Experiment breaks theory.
Numbers decide.
You are ready for questions.

“Respected teacher and friends, thank you for listening. We are ready for questions.”

References, Sources and Image Credits — Trace Every Claim

ALL MEMBERS 01–10

- Teaching source (primary): the 8-page Detailed Teach-the-Teacher Script — all dates, numbers, equations, and Q&A above come from it.
- Supplementary textbook context, flagged in notes: Rydberg R, Planck h, Heisenberg 1927 / Schrödinger 1926 framing, hc — verify against your course text.
- Images (Wikimedia Commons, accessed 2026-10-15): Thomson pre-1915 photo (PD); Rutherford, Bain News Service via LoC (PD-US); Bohr c.1922 (PD); Geiger–Marsden 1913 diagram (PD); H spectrum, Umop503 (CC0); orbital densities, PoorLeno (PD); CRT photo, Hebels (PD, illustrative).
- Diagrams labelled custom schematic were drawn natively in PowerPoint for accuracy — not to scale unless stated; orbit rings are always historical-model labelled.
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– PRIMARY: 8-page Teach-the-Teacher Script — all dates, numbers, Q&A.

– SUPPL: Rydberg R, Planck h, 1926–27 framing, hc — check textbook.

– IMAGES: Commons PD/CC0 — Thomson / Rutherford (LoC) / Bohr / Geiger–Marsden / Umop503 / PoorLeno / Hebels.

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